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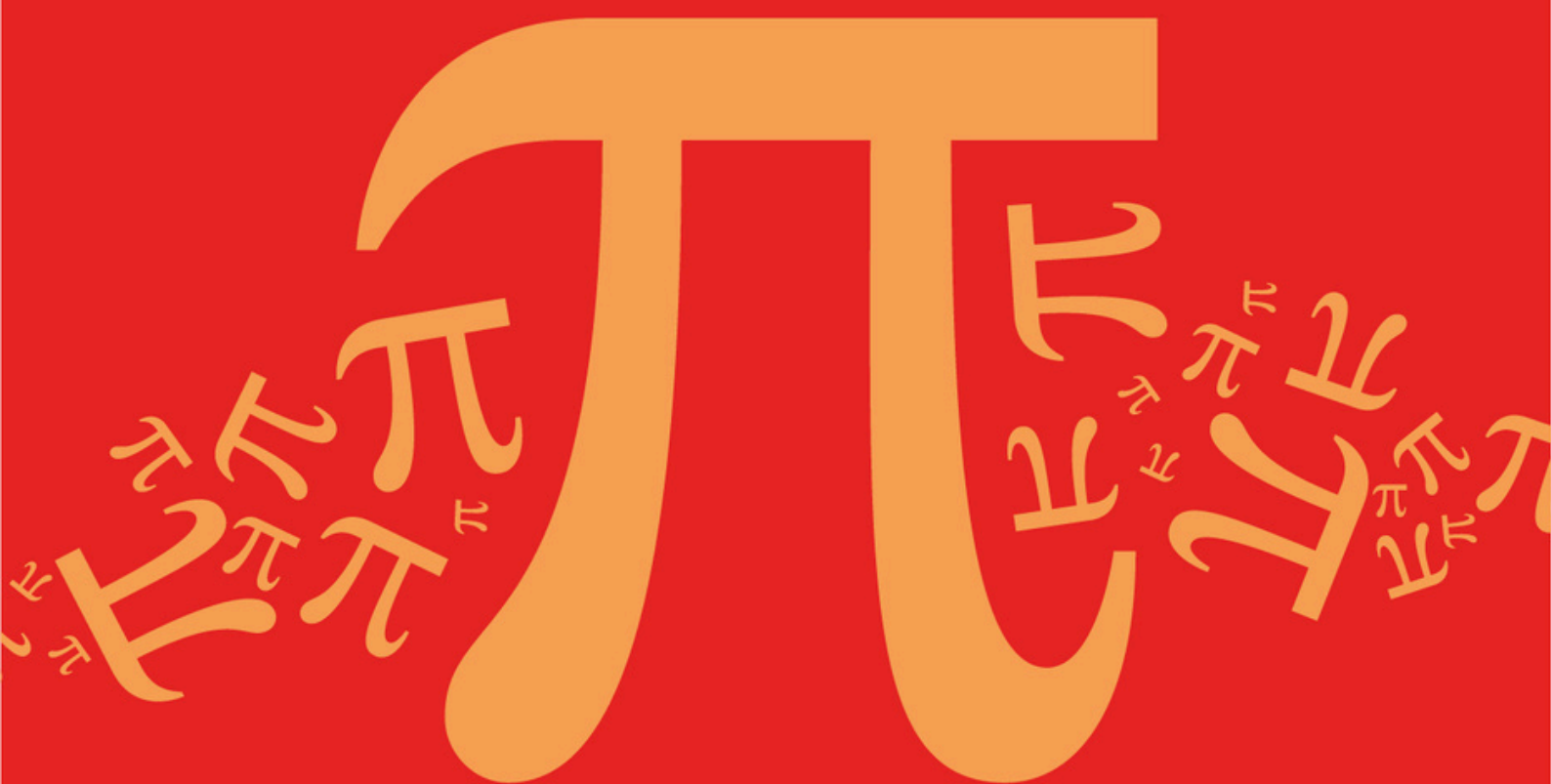
Practice questions created by actual
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Detailed mark scheme

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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C11: Diagonalisation

Mark Scheme

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Mark Scheme

Topic

C: Matrices

Sub topic

C11: Diagonalisation

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms appropriate equation using $\mathbf{M}\lambda = \lambda\mathbf{v}$	AO1.1a	M1	$\begin{bmatrix} -1 & 2 & -1 \\ 2 & 2 & -2 \\ -1 & -2 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 4 \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ $\begin{aligned} -5a + 2b - 2c &= 0 \\ 2a - 2b - 2c &= 0 \\ -a - 2b - 5c &= 0 \\ 3a + 3c &= 0 \end{aligned}$ $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ eigenvector is
	Eliminates one variable	AO1.1a	M1	
	Deduces a correct eigenvector	AO2.2a	A1	
(b)	Forms the characteristic equation of $\mathbf{M}$	AO3.1a	M1	$\begin{vmatrix} -1-\lambda & 2 & -1 \\ 2 & 2-\lambda & -2 \\ -1 & -2 & -1-\lambda \end{vmatrix} = 0$ $(-1-\lambda)[(2-\lambda)(-1-\lambda) - 4]$ $-2(-2-2\lambda-2) - 1(-4+2-\lambda) = 0$ $-\lambda^3 + 12\lambda + 16 = 0$ $(4-\lambda)(\lambda^2 + 4\lambda + 4) = 0$ $-(\lambda+2)(\lambda-4)(\lambda+2) = 0$ Eigenvalues are 4, -2, -2 $\begin{bmatrix} -1 & 2 & -1 \\ 2 & 2 & -2 \\ -1 & -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -2 \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $x + 2y - z = 0$
	Obtains the correct characteristic equation – unsimplified	AO1.1b	A1	
	Obtains roots and identifies them as eigenvalues for ‘their’ characteristic equation	AO1.1b	A1F	
	Forms an appropriate matrix equation using the eigenvalue -2 FT ‘their’ eigenvalue	AO3.1a	M1	
	Expands and simplifies to obtain a single equation in x, y and z FT ‘their’ matrix equation provided both M1 marks have been awarded	AO1.1b	A1F	
	Correctly deduces two linearly independent eigenvectors CAO	AO2.2a	A1	
	Correctly identifies that the matrix $\mathbf{D}$ must include 4	AO1.2	B1F	

and 'their' other eigenvalue(s)			$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$
Correctly identifies the corresponding U matrix from 'their' eigenvectors	AO1.1b	A1F	$\mathbf{D} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ $\mathbf{U} = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix}$
Total 11 marks			

Q2.

(a)
$$\begin{bmatrix} -a & 0 & a \\ 0 & 6 & 0 \\ a & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \\ 0 \end{bmatrix}$$

M1

$\Rightarrow \lambda_1 = 6$

A1

(b)
$$\begin{bmatrix} -a & 0 & a \\ 0 & 6 & 0 \\ a & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} a \\ 0 \\ a+4 \end{bmatrix}$$

M1

$$\begin{bmatrix} a \\ 0 \\ a+4 \end{bmatrix} = \begin{bmatrix} \lambda_2 \\ 0 \\ 2\lambda_2 \end{bmatrix}$$

i component $\Rightarrow a = \lambda_2 \quad \otimes$

k component $\Rightarrow a + 4 = 2\lambda_2$

using $\otimes$, $a + 4 = 2a$
Eliminating λ_2

m1

$4 = a$

Value of a obtained

A1

(c) Let $\mathbf{v}_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\begin{bmatrix} -4 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -6 \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Substitute 'their value of a' and attempt to get a system of equations.

M1

$$\Rightarrow -4x + 4z = -6x \Rightarrow x + 2z = 0$$

$$6y = -6y \Rightarrow y = 0$$

$$[4x + 2z = -6z \Rightarrow x + 2z = 0]$$

Both equations "correct" FT their a

A1F

$$\mathbf{v}_3 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \text{ (or equivalent)}$$

A1cao

Alternative:

$$\begin{bmatrix} -4 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = -6 \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 0 & 4 \\ 0 & 12 & 0 \\ 4 & 0 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$r_3 \rightarrow r_3 - 2r_1 \quad \begin{bmatrix} 2 & 0 & 4 \\ 0 & 12 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Row operations

(M1)

$$\Rightarrow y = 0 \quad x = -2z$$

"correct" FT their a

(A1F)

$$\mathbf{v}_3 = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \text{ (or equivalent)}$$

(A1cao)

(3)

(d) $\mathbf{D} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -6 \end{bmatrix}$

Diagonal matrix using -6 and "their 4" and "their 6"

B1F

$\mathbf{U} = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

*FT their non-zero $\mathbf{v}_3$ in $\mathbf{U}$
 $\mathbf{U}$ correct and corresponding to $\mathbf{D}$*

A1cao

3

[11]

Q3.

- (a) (i) Characteristic eqn $\lambda^2 - 9\lambda + 14 = 0$
M1A0 if num error(s) made

M1A1

$\lambda = 2, 7$

A1

Substituting back for at least one eval

for $\lambda = 2$, $-x + 3y = 0$ or

for $\lambda = 7$, $-2x + y = 0$

m1

evecs $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
or non-zero multiples

A1A1

6

(ii) $\mathbf{U} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$

*Columns of $\mathbf{U}$ and $\mathbf{D}$ are interchangeable,
 but must match;
 ft wrong answers in (i)*

B1FB1F

$$\mathbf{U}^{-1} = \frac{1}{\det \mathbf{U}} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}$$

adjoint matrix;
ft incorrect $\mathbf{U}$ (provided $\det \neq 0$)

B1F

B1F

4

(b) (i) evals of $\mathbf{M}^3$ are λ^3, μ^3

since $\mathbf{M}^3 = \mathbf{U} \mathbf{D}^3 \mathbf{U}^{-1}$

B1

E1

2

(ii) evects of $\mathbf{M}^3$ are $\mathbf{v}_1$ and $\mathbf{v}_2$

B1

1

[13]

Q4.

(a) (i) Evals $\lambda = 27$
Both

1

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B1

Evecs $\begin{pmatrix} \alpha \\ 4 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} \beta \\ -1 \\ 3 \end{pmatrix}$

Correctly matched up with evals
(look out for $\lambda_1, \mathbf{v}_1$ notations)

B1B1B1

4

(ii) $y = -3x$

Ft $4y = x$ if evects mis-matched

B1ft

from $\lambda = 1$

Must say why they have chosen this one

B1

2

(b) $U^{-1} = \frac{1}{13} \begin{bmatrix} 3 & 1 \\ -1 & 4 \end{bmatrix}$
Det; mtx

B1B1

$M = U D U^{-1}$
*Including attempt to multiply
 (at least $U D \dots$)*

M1

$$= \frac{1}{13} \begin{bmatrix} 4 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 27 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 4 \end{bmatrix}$$

$$= \frac{1}{13} \begin{bmatrix} 4 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 81 & 27 \\ -1 & 4 \end{bmatrix}$$

*Ft incorrect/missing U^{-1} for one product;
 ignore missing $\frac{1}{13}$ until the end*

A1

or $\frac{1}{13} \begin{bmatrix} 108 & -1 \\ 27 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 4 \end{bmatrix}$
 $= \begin{bmatrix} 25 & 8 \\ 6 & 3 \end{bmatrix}$

CAO

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5

(c) $M^n = U D^n U^{-1}$
Including attempt to multiply

M1

$$D^n = \begin{bmatrix} 27^n & 0 \\ 0 & 1 \end{bmatrix}$$

B1

$$M^n(1, 1) = \frac{1}{13} (12 \times 27^n + 1)$$

A1

So $4 \times 3 \times 3^{3n} + 1 = 4 \times 3^{3n+1} + 1$ div. by 13
*Legitimately so from their working, from
 fact that the element is an integer*

E1

Since $\mathbf{M}$ has all integer elements, each element of $\mathbf{M}^n$ is an integer also

Explaining why it must be an integer

E1

5

[16]

Q5.

(a) (i) $\mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix} \quad \mathbf{U} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}$
 $\mathbf{D}, \mathbf{U}$ (alt. choices ok)

B1B1

$$\mathbf{U}^{-1} = \frac{1}{2} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$$

ft 1st B1 provided $\det \neq 0$

B1

ft 2nd B1 in non-trivial cases

B1

4

(ii) $\mathbf{M} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix}$
Some attempt at mtx. multn.

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M1

$$= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 12 & -3 \\ 6 & -3 \end{bmatrix} \text{ or}$$

First multn. correct ft

A1

$$\begin{aligned} & \frac{1}{2} \begin{bmatrix} 3 & -3 \\ 6 & -12 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 9 & -3 \\ 24 & -9 \end{bmatrix} \end{aligned}$$

Ft missing $\frac{1}{2}$ only

A1

3

(b) (i) When n even, $\mathbf{D}^n = \begin{bmatrix} 3^n & 0 \\ 0 & 3^n \end{bmatrix}$

Incl. use in mtx. multn. of form $\mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$

M1

$$\mathbf{M}^n = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \cdot 3^n & -3^n \\ -2 \cdot 3^n & 3^n \end{bmatrix} \text{ or } \frac{1}{2} \begin{bmatrix} 3^n & 3^n \\ 2 \cdot 3^n & 4 \cdot 3^n \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \text{ correct}$$

Correct ft

A1

Showing $\mathbf{M}^n = 3^n \mathbf{I}$ legitimately

A1

3

(ii) When n odd, $\mathbf{D}^n = \begin{bmatrix} 3^n & 0 \\ 0 & -3^n \end{bmatrix}$
 Incl. use in mtx. multn. of form $\mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$

M1

$$\mathbf{M}^n = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 4 \cdot 3^n & -3^n \\ 2 \cdot 3^n & -3^n \end{bmatrix} \text{ or } \frac{1}{2} \begin{bmatrix} 3^n & -3^n \\ 2 \cdot 3^n & -4 \cdot 3^n \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -2 & 1 \end{bmatrix} \text{ correct}$$

Correct ft

A1

Showing $\mathbf{M}^n = 3^{n-1} \mathbf{M}$ legitimately

A1

3

[13]

Q6.

(a) $\mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix},$

B1B1

$$\mathbf{U}^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

ft $\mathbf{U}^{-1}$

B1

3

(b) $T^n = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2^n & 0 \\ 0 & 2^n \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$

For D^n with n even

B1

For use of $U^{-1} D^n U$ form

M1

$$= \begin{bmatrix} 2^n & 2 \times 2^n \\ 2^n & 3 \times 2^n \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

m1A1

or $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 \times 2^n & -2 \times 2^n \\ -2^n & 2^n \end{bmatrix}$
 $= 2^n \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Shown legitimately

A1

5

Alternative:

$$D^n = \begin{bmatrix} 2^n & 0 \\ 0 & 2^n \end{bmatrix}$$

For D^n with n even

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$$T^n = U (2^n I) U^{-1}$$

(M1)

$$= 2^n (U I U^{-1})$$

(m2)

$$= 2^n I$$

Allow $\equiv$ forms such as $3 \cdot 2^n - 2^{n+1}$

(A1)

(5)

[8]

Q7.

(a) $y = 0$ (or "x-axis") and $y = x$

$$\text{or } \mathbf{r} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{r} = b \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

B1,B1

$y = 0$ is a line of invariant points since $\lambda = 1$

Allow if proven from $(x', y') = (x, y)$ or ft from their line corresponding to $\lambda = 1$

B1

3

(b) (i) $\mathbf{D} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $\mathbf{U} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$,
ft $\mathbf{U}$ from $\mathbf{D}$

B1,B1

2

(ii) $\mathbf{U}^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
ft from $\mathbf{U}$ (provided non-singular)

B1

$\mathbf{M} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
Attempt

M1

$= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ or $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$
ft first multiplication

A1

$= \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$

CAO

$NMS \Rightarrow 0$

A1

4

(iii) $\mathbf{D}^x = \begin{bmatrix} 1 & 0 \\ 0 & 2^x \end{bmatrix}$

Noted or used

B1

$\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$

Used; must actually do some multiplying

M1

$$= \begin{bmatrix} 1 & 2^x - 1 \\ 0 & 2^x \end{bmatrix}$$

A1

3

[12]

Q8.

(a) Char. Eqn. is $\lambda^2 - 5\lambda - 6 = 0$

B1

Solving $\Rightarrow \lambda = -1$ or 6

M1 A1

Substⁿ. either λ back

M1

$\lambda = -1 \Rightarrow x + y = 0 \Rightarrow$ evecs. $\alpha \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
 Any non-zero α

A1

$\lambda = 6 \Rightarrow 5x - 2y = 0 \Rightarrow$ evecs. $\beta \begin{bmatrix} 2 \\ 5 \end{bmatrix}$
 Any non-zero β

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A1

6

(b) (i) $\mathbf{D} = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix}$ $\mathbf{U} = \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix}$
 ft evals.

B1F

ft evecs. (must correspond to their evals.)

B1F

2

(ii) $\mathbf{U}^{-1} = \frac{1}{7} \begin{bmatrix} 5 & -2 \\ 1 & 1 \end{bmatrix}$
 ft their $\mathbf{U}$ (provided non-singular)

B1F

1

(iii) $\mathbf{X}^5 = \mathbf{U} \mathbf{D}^5 \mathbf{U}^{-1}$

M1

$$= \frac{1}{7} \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 6^5 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 1 & 1 \end{bmatrix}$$

Use of correct D^5 (ft)
N.B. $6^5 = 7776$

B1F

$$= \begin{bmatrix} 2221 & 2222 \\ 5555 & 5554 \end{bmatrix}$$

A1

3

[12]

Q9.

(a) (i) Char. Equation is

M1

$$\lambda^2 - 0\lambda + \{-a^2 - (b^2 - a^2)\} = 0$$

Correct to at least here

A1

i.e. $\lambda^2 - b^2 = 0$ and $\lambda = \pm b$

A1

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3

(ii) $\lambda = b \Rightarrow (a - b)x + (a + b)y = 0$

M1

$$\Rightarrow y = \frac{b-a}{b+a}x \Rightarrow \text{eigen vectors } \alpha \begin{bmatrix} b+a \\ b-a \end{bmatrix}$$

AG

A1

2

(iii) $\lambda = -b \Rightarrow (a + b)x + (a + b)y = 0$ (etc.)

M1

$$\Rightarrow y = -x \Rightarrow \text{eigen vectors. } \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

A1

2

(b) $\mathbf{D} = \begin{bmatrix} b & 0 \\ 0 & -b \end{bmatrix}, \mathbf{U} = \begin{bmatrix} b+a & 1 \\ b-a & -1 \end{bmatrix}$

B1 B1

and $\mathbf{U}^{-1} = \frac{-1}{2b} \begin{bmatrix} -1 & -1 \\ a-b & a+b \end{bmatrix}$
ft

B1 ✓

$\mathbf{D}^{11} = \begin{bmatrix} b^{11} & 0 \\ 0 & -b^{11} \end{bmatrix}$

B1

$\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$ used

M1

$\mathbf{M}^{11} = \frac{1}{2} b^{10} \begin{bmatrix} b+a & 1 \\ b-a & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ a-b & a+b \end{bmatrix}$

or $\frac{1}{2} b^{10} \begin{bmatrix} b+a & -1 \\ b-a & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ b-a & -a-b \end{bmatrix}$

A1

$= \frac{1}{2} b^{10} \begin{bmatrix} 2a & 2(a+b) \\ 2(b-a) & -2a \end{bmatrix} = b^{10} \mathbf{M}$

AG

A1

7

Alternative to (a)(ii)

$\mathbf{M} \begin{bmatrix} b+a \\ b-a \end{bmatrix} = \begin{bmatrix} ab+a^2+b^2-a^2 \\ b^2-a^2-ab+a^2 \end{bmatrix} = b \begin{bmatrix} b+a \\ b-a \end{bmatrix}$

M1

A1

Alternative to (b)

NB $\mathbf{D}^{11} = b^{10} \mathbf{D}$

B1

Then $\mathbf{M}^{11} = \mathbf{U} \mathbf{D}^{11} \mathbf{U}^{-1}$

M2 A1

$$= b^{10} \mathbf{U} \mathbf{D} \mathbf{U}^{-1} = b^{10} \mathbf{M}$$

M2 A1

[14]

Q10.

(a) $\begin{bmatrix} 2 & 7 \\ 4 & k \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 4+k \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ when $k = 5$,
 $\mathbf{M} \times$ given evec.

M1A1

(b) Char. Eqn. is $\lambda^2 - 7\lambda - 18 = 0$
 $(\lambda - 9)(\lambda + 2) = 0$ and 2nd eval. is -2

A1

3

M1A1

A1

OR $\det \mathbf{M} = \lambda_1 \lambda_2 \Rightarrow -18 = 9\lambda_2$
 $\Rightarrow \lambda_2 = -2$

Or via trace $\mathbf{M} = \lambda_1 + \lambda_2$

Substⁿ. $\lambda = -2 \Rightarrow 4x + 7y = 0$

M1

$\Rightarrow$ evec. $\begin{bmatrix} 7 \\ -4 \end{bmatrix}$

A1

5

(c) $\mathbf{D} = \begin{bmatrix} -2 & 0 \\ 0 & 9 \end{bmatrix}$, $\mathbf{U} = \begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix}$
 ft (alternatives possible)

B1B1

2

(d) $\mathbf{U}^{-1} = \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix}$
 ft non-trivial $\mathbf{U}$'s

B1

1

(e)
$$\mathbf{M}^{2n} = \begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} (-2)^{2n} & 0 \\ 0 & (9)^{2n} \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix} = \begin{bmatrix} 7 \times 4^n & 81^n \\ -4^{n+1} & 81^n \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 1 & -1 \\ 4 & 7 \end{bmatrix}$$

For D^{2n}

B1

or

$$\begin{bmatrix} 7 & 1 \\ -4 & 1 \end{bmatrix} \cdot \frac{1}{11} \begin{bmatrix} 4^n & -4^n \\ 4 \times 81^n & 7 \times 81^n \end{bmatrix}$$

CAO any correct form

M1

A1

Thus $a = \frac{1}{11} \{7 \times 4^n + 4 \times 81^n\}$

i.e. $p = \frac{7}{11}, q = \frac{4}{11}$

A1

4

In its original form, the question asked for the following conclusion to be made:

Since a is an integer, and $\text{hcf}(4, 11) = 1$,
 $7 \times 4^{n-1} + 81^n$ is a multiple of 11

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[15]

Examiner reports

Q2.

Many students were able to show their understanding of eigenvalues and eigenvectors, scoring full marks in the process. They all approached the question by using the definition of eigenvalues and eigenvectors: $\mathbf{M}\mathbf{v} = \lambda\mathbf{v}$. The unsuccessful students tried in vain to use the characteristic equations getting lost in pages of algebra. The $\mathbf{UDU}^{-1}$ form was well understood by almost all students.

Q3.

Candidates generally showed a very good grasp of eigenvalues and eigenvectors in part (a)(i), though some were confused when forming the eigenvectors. The responses to part (a)(ii) were almost equally successful, but some faltered in tackling the inverse of a 2 by 2 matrix, either failing to include the reciprocal of the determinant or incorrectly manipulating the elements of the matrix to form the adjoint matrix.

In part (b)(i), many candidates earned partial credit, either by giving the eigenvalues but failing to show the connection with the diagonalised form, or by doing some good work with the diagonalised form and then failing to state clearly what the eigenvalues were. Part (b)(ii) was answered correctly in many cases but a common error was to write down the meaningless answers ' $\mathbf{v}_1^3$ ' and ' $\mathbf{v}_2^3$ '. Some candidates used the matrix from part (a) in their answers to part (b), but this was accepted as they could still show whether they understood what was needed in part (b).

Q4.

This question was handled very well indeed in its technical aspects – writing down eigenvalues and eigenvectors; finding a 2×2 inverse matrix; determining $\mathbf{M}$, and the accompanying matrix multiplications. So a lot of candidates scored lots of marks on this question. However, there were very few candidates indeed who managed to score more than 12 or 13, due to the explanations that were being looked for. At its simplest level, this began in part (a)(i), where candidates were required to write down “the eigenvalues and **corresponding** eigenvectors ...” – in other words, to say which went with which. Examiners did not accept a fortuitously ordered separate pair of each (with a 50-50 chance of a correct pairing). In part (a)(ii), we also wanted to know **why** they had chosen one of the two possible lines as a line of invariant points (as opposed to just an invariant line).

Part (c) had been designed to require thought of the candidates, so the difficulties that arose here were not unexpected. There were two marks for explaining why the given expression was divisible by 13. The first was awarded to anyone who noted that the 1-1 element of $\mathbf{M}^n$ is an integer; the second, subtler, one was for the explanation of why it had to be an integer. Disappointingly few candidates got both marks.

Q5.

The more confident candidates found this question to be a source of easy marks. The real difficulties arose in part (b) due to a widespread inability to use brackets and deal with minus signs. When candidates wrote -3^n when they actually meant $(-3)^n$, and it was necessary to read on to see what they eventually did with the terms involving powers of (-3) . The standard approach appears in the mark scheme, though the alternative algebraic ones also appeared quite often, and these were much more pleasing to the eye. For n

even, noting that $\mathbf{D}^n = 3^n \mathbf{I}$, we have $\mathbf{M}^n = 3^n \mathbf{U} \mathbf{I} \mathbf{U}^{-1}$, which gives the required result both obviously and quickly. However, a small number attempted this approach, but wrote something along the lines of $\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1} = \mathbf{D}^n \mathbf{U} \mathbf{U}^{-1} = \mathbf{D}^n$ first. This is wrong and

scored 0/3. For n odd, a similar 'shortcut' arises via $\mathbf{M}^n = 3^n \mathbf{U} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{U}^{-1}$. However, a very small number of candidates are to be highly commended for the following insightful, approach: if n is odd, then $n - 1$ is even, so that $\mathbf{M}^n = \mathbf{M}^{n-1} \mathbf{M} = (3^{n-1} \mathbf{I}) (\mathbf{M})$ using the result of part (b)(i) $= 3^{n-1} \mathbf{M}$, as required.

Q6.

This question was actually quite straightforward. However, apart from the final parts of Question 7, it managed to present the biggest obstacle on the paper. Surprisingly, the greatest difficulty arose when candidates were required to turn $(-2)^n$ into 2^n , in the case when n is even, when considering the matrix $\mathbf{D}^n$. Many failed to manage this at all. A disappointingly large number of candidates left it until well into their working and many such candidates then failed to cope with the complicated expressions they were trying to deal with. Those that did so at the outset found themselves with an alternative and simple route through the problem available to them, since it follows that $\mathbf{D}^n = 2^n \mathbf{I}$; it is now straightforward to re-write $\mathbf{T}^n$ and to note that $f(n) = 2^n$.

Q7.

Most difficulties with this question arose in part (a), where candidates had to consider the meaning and relevance of the information given to them in the question. The most commonly lost mark was in the widespread failure to point out that the line of invariant points was flagged up by the fact that the eigenvalue corresponding to the x -axis was 1. In general, most other marks lost on this question arose as a result of minor slips and oversights in the working. A few candidates didn't seem to know how to find $\mathbf{M}^n$.

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Q8.

This question was handled very confidently by most candidates, and apart from a small number of candidates who clearly deduced the matrix $\mathbf{X}^5$ from a calculator, despite the question's injunction to use the previous results, marks of 10, 11 or 12 were very common indeed.

Q9.

The algebraic nature of this question caused many problems, particularly in part (b). For part (a), most candidates had little difficulty in following the standard routine for finding eigenvectors, although many caused themselves problems by failing to simplify the

second eigenvector from $\begin{bmatrix} a+b \\ -b-a \end{bmatrix}$ to $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Any correct form was acceptable for the eigenvectors, and if mistakes occurred the candidates could still carry on and earn follow-through marks.

As mentioned above, many candidates did not make any serious attempt at part (b). Of those who did try this part, the overwhelming majority made life particularly hard for themselves by failing to do some fairly simple algebraic processes, such as factorising.

Thus, the product
$$\begin{bmatrix} b+a & 1 \\ b-a & -1 \end{bmatrix} \begin{bmatrix} b^{11} & 0 \\ 0 & -b^{11} \end{bmatrix} \begin{bmatrix} \frac{1}{2b} & \frac{1}{2b} \\ -\frac{a-b}{2b} & \frac{a+b}{2b} \end{bmatrix}$$
 could have been made to look so much easier by taking out the factors of b^{11} and $\frac{1}{2b}$ to get
$$\frac{1}{2} b^{10} \begin{bmatrix} b+a & 1 \\ b-a & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ b-a & -a-b \end{bmatrix}.$$

It was also unfortunate that so many candidates were unable to find the inverse matrix for their choice of U . Some candidates saw that there was no need to find U and U^{-1} at all. The fact that $D^{11} = b^{10} D$ led them to the reasoning: $M^{11} = U D^{11} U^{-1} = U b^{10} D U^{-1} = b^{10} (U D U^{-1}) = b^{10} M$, though not always expressed quite so concisely.

Q10.

Most candidates managed to score some marks here, Even those picking up 2 or 3 of the marks available here did so retrospectively. The most common error in part (b) was to fail to turn the equation $4x + 7y = 0$, gained from the second eigenvalue of -2 , into a correct eigenvector. Also, the factor of $1/\det U$ was often overlooked. Part (e) was seldom attempted. It was, however, very pleasing to see some completely correct conclusions to the paper.